

ERES Cryptographic Stack Primitive v1.3

Symbols · Characters · Talonics — A Resonance-Keyed Semantic Superposition Class

Author: Joseph Allen Sprute (ERES Maestro)

Affiliation: ERES Institute for New Age Cybernetics, Bella Vista, AR 72714

ORCID: 0000-0001-9946-3221

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Abstract

This paper formalizes the ERES Cryptographic Stack Primitive at version 1.3, locking the resonance-keyed semantic superposition primitive class as the foundational cryptographic substrate for ERES Institute architectures. The primitive is expressed by the canonical relation **(CBGMODD × FAVORS) + BERA = RATE**, where seven RATE dimensions (R_1 – R_7) are derived from a three-tier semantic stack: Symbols (atomic glyph layer), Characters (compositional identity layer), and Talonics (governance-state layer, $18 \times 7 = 126$ states across 12 parts and 452 elements). Version 1.3 introduces the **VEILED** state as a principled replacement for the conventional bottom value ($\perp$), preserving epistemic incompleteness without collapse into falsity. It additionally formalizes **Trifurcated Math** as the third pedagogical isomorph of Triune Math #2, completing the family of canonical keys: (1) $C = R \times P / M$; (2) $M \times E + C = R$; (3) $REAL = (E \cdot M \cdot R) / (T \cdot S)$. The primitive is the attestation substrate for SCALULAR and the upstream dependency for EAAP v1.2.

Keywords: cryptographic primitive, semantic superposition, Talonics, VEILED state, Triune Math, Trifurcated Math, BERA, RATE, resonance keying, ERES Institute.

1. Introduction

The ERES Cryptographic Stack Primitive is not a hash function, not a signature scheme, and not a key-exchange protocol. It is the *semantic substrate* on which those operational primitives are layered. Conventional cryptography assumes meaning is exogenous to the cryptographic object — a signed bit-string is a signed bit-string, regardless of what the bits represent. The ERES Stack Primitive inverts that assumption: meaning is endogenous, carried by a three-tier semantic stack whose composition rules generate cryptographically usable structure. Versions 1.0 through 1.2 established the layered stack and the operational relation. Version 1.3 closes three open items: (i) the VEILED state, (ii) Trifurcated Math, and (iii) the formal class

definition of resonance-keyed semantic superposition.

2. The Three-Tier Stack

2.1 Symbols. The atomic glyph layer. Symbols carry no compositional commitments; they are addressable, enumerable, and distinguishable. The Symbol layer is closed under identity but not under combination.

2.2 Characters. The compositional identity layer. Characters are constructed from Symbols under named composition rules and acquire stable identity across contexts. A Character is the smallest unit at which an attestable claim can be made.

2.3 Talonics. The governance-state layer. The Talonics Symbol Matrix consists of 452 elements organized into 12 parts, generating $18 \times 7 = 126$ governance states. Each Talonic state encodes a compositional commitment about how Characters may be combined under a given governance regime. Talonics are the layer at which cryptographic operations acquire *normative* (not merely syntactic) force.

3. The Canonical Relation

The operational relation of the stack is:

$$(\text{CBGMODD} \times \text{FAVORS}) + \text{BERA} = \text{RATE}$$

Where:

- **CBGMODD** — the seven-seat Family Stewardship Council membership vector (Caretakers, Beneficiaries, Guardians, Mediators, Officers, Directors, Delegates).
- **FAVORS** — the multidimensional compatibility lattice between participating roles and proposed actions.
- **BERA** — the Behavioral and Environmental Resonance Audit, a measurement framework producing scalar and vector outputs across ARI, ERI, RHC, and RCI dimensions.
- **RATE** — the seven-dimensional output (R_1 – R_7), each dimension keyed to a distinct attestation surface.

3.1 The seven RATE dimensions:

- **R_1 — Receipt:** the cryptographic acknowledgment that the relation has executed.
- **R_2 — Reach:** the scope of CBGMODD seats whose authority is engaged.
- **R_3 — Resonance:** the BERA-derived scalar at moment of attestation.
- **R_4 — Resilience:** the projected stability of the attestation under adversarial perturbation.
- **R_5 — Reciprocity:** the FAVORS-derived obligation vector returned to participating roles.
- **R_6 — Record:** the immutable trace deposited to the attestation substrate.
- **R_7 — Right:** the normative claim — under the active Talonics state — that the attestation confers.

4. The VEILED State

In conventional logic, $\perp$ denotes falsity, contradiction, or the empty model. In cryptographic protocols, $\perp$ frequently overloads to denote "no value," "abort," or "rejection." This overloading is corrosive in attestation contexts: an absent attestation, a withheld attestation, and a denied attestation are not the same epistemic event.

Version 1.3 introduces **VEILED** as a principled replacement for $\perp$ in the ERES Stack Primitive. A VEILED state asserts: *the Right (R_7) is not currently displayed at this attestation surface, but*

the underlying relation has not collapsed. VEILED preserves epistemic incompleteness without forcing it into falsity. It is recoverable under proper credentialing; $\perp$ is not. VEILED is the formal accommodation, at the cryptographic layer, of the IDIPITIS principle that not all knowledge is concurrently disclosed knowledge.

5. Trifurcated Math as Third Pedagogical Isomorph

Triune Math is taught in three pedagogical isomorphs:

- **Key 1 — Cybernetic form:** $C = R \times P / M$ (Cybernetics = Resource $\times$ Purpose $\div$ Method).
- **Key 2 — Physical form:** $M \times E + C = R$ (Matter $\times$ Energy + Constant = Reason).
- **Key 3 — Resonance form:** $REAL = (E \cdot M \cdot R) / (T \cdot S)$ (Energy $\cdot$ Matter $\cdot$ Resonance over Time $\cdot$ Space; no repeated letters).

Trifurcated Math is the third pedagogical isomorph of Key 2, presenting $M \times E + C = R$ as a branching structure that exposes the three contributions (Matter, Energy, Constant) as separable but co-determinative inputs to Reason. Trifurcated Math is the bridge between Triune Math's algebraic compactness and the Talonics layer's compositional richness; it is what makes the cryptographic primitive teachable to non-specialist participants.

6. The Primitive Class

The **resonance-keyed semantic superposition** primitive class is defined as the family of cryptographic objects O such that:

1. O is constructed over a three-tier semantic stack (Symbols / Characters / Talonics).
2. O 's operational relation is of the form (council $\times$ compatibility) + audit = output, with output decomposed into seven attestation dimensions.
3. O admits a VEILED state distinct from rejection.
4. O 's pedagogical exposition is consistent with the three Triune Math isomorphs and Trifurcated Math.

Any cryptographic object meeting these four criteria is a member of the class and inherits the attestation guarantees of the ERES Stack Primitive.

7. Position in the ERES Stack

The Stack Primitive is the upstream dependency for:

- **EAAP v1.2** — operational attestation protocol (see ERES-EAAP-2026-001-v1.2, slot 432).
- **SCALULAR** — Scalable Certification Architecture for Lifelong Universal Learning and Adaptive Resilience.
- **Meritcoin** — Proof-of-Resonance valuation, where R_3 feeds value derivation directly.

It is downstream of: the 72 Key Domains of CyberRAVE, the IDIPITIS protocol, and the BERA measurement framework.

8. Conclusion

Version 1.3 closes the foundational layer. With the VEILED state principled, Trifurcated Math seated as the third pedagogical isomorph of Key 2, and the primitive class formally defined,

downstream protocols (EAAP, SCALULAR, Meritcoin) can cite a stable upstream rather than re-derive primitive structure. Future revisions are anticipated to address quantum-resistant lifting and inter-stack composition with non-ERES attestation substrates.

References

1. Sprute, J. A. (2026). *ERES UBIMIA Manual v1.1*. ERES Institute. (CCAL v2.1.)
2. Sprute, J. A. (2026). *ERES Talonics Symbol Matrix and the Talonics Commons License (TCL v1.0)*. ERES Institute.
3. Sprute, J. A. (2026). *DOFA Whitepaper Version C*. ERES-DOFA-WP-2026-001-C.
4. Sprute, J. A. (2026). *Berggruen Prize Essay: Intelligent Design as Civilizational Option*.
5. Sprute, J. A. (2026). *ERES Hague Filing*. ERES-HAGUE-2026-001.

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